# A Comparative Evaluation of AI-Powered Literature Discovery Tools Across Disciplines

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The rapid growth of academic publications has made literature discovery increasingly difficult for students and researchers. Conventional keyword-based academic databases remain essential, but artificial intelligence (AI) has introduced new approaches to finding, organizing, and connecting scholarly literature. Tools such as Semantic Scholar, Elicit, ResearchRabbit, Connected Papers, and Litmaps use different combinations of semantic search, citation analysis, machine learning, and knowledge-graph techniques. This project presents a comparative evaluation of these AI-powered literature discovery tools across four academic disciplines: computer science, biomedical science, social science, and humanities. The evaluation framework considers relevance, coverage, discovery of influential and recent publications, efficiency, usability, and transparency. An illustrative evaluation suggests that no single tool consistently performs best across all disciplines and tasks. Semantic Scholar and Elicit are particularly useful for semantic and question-oriented discovery, while ResearchRabbit, Connected Papers, and Litmaps provide strong support for citation-network exploration. The study argues that AI literature-discovery tools are most effective when used as complementary technologies alongside conventional scholarly databases and researcher judgment.

CCS Concepts: • **Information systems** → **Information retrieval**; *Digital libraries and archives*; • **Computing methodologies** → *Artificial intelligence.*

Additional Key Words and Phrases: artificial intelligence, literature discovery, academic search, Semantic Scholar, Elicit, ResearchRabbit, Connected Papers, Litmaps, scholarly communication

## 1 Introduction

The volume of scientific and scholarly literature has increased substantially over the past several decades. Researchers must therefore identify relevant publications from increasingly large and complex information environments. Traditional academic databases provide powerful keyword, author, subject, and citation searches, but discovering less obvious connections between publications can still be challenging.

Recent advances in artificial intelligence have changed the way researchers can approach literature discovery. AI-powered systems can use semantic representations, citation relationships, machine-learning models, and natural-language processing to recommend papers that may not contain exactly the same keywords as a user's query. Several platforms now combine these capabilities with features designed specifically for literature reviews.

Semantic Scholar, for example, provides AI-assisted scholarly search and citation information. Elicit focuses on research questions and evidence extraction. ResearchRabbit and Connected Papers emphasize relationships between publications, while Litmaps allows researchers to visualize and monitor citation networks. Recent scholarship identifies these systems as increasingly important components of contemporary literature-review workflows [2, 6].

However, the existence of AI capabilities does not necessarily mean that all tools perform equally well. Academic disciplines differ in terminology, publication patterns, citation practices, and database coverage. A tool that works particularly well for biomedical research may not provide the same level of coverage for humanities scholarship.

This project therefore examines the following research question:

> How effectively do AI-powered literature discovery tools support scholarly literature discovery across different academic disciplines?

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The study also addresses three secondary questions:

(1) Which tools are most effective for identifying relevant literature?
(2) How does tool performance differ between academic disciplines?
(3) Can a combination of AI-powered tools provide better discovery support than reliance on a single platform?

## 2 Literature Review

### 2.1 AI and Scholarly Literature Discovery

Artificial intelligence has increasingly been incorporated into academic information systems. Machine-learning approaches can identify semantic relationships between documents, classify publications, recommend related papers, and assist researchers in navigating large scholarly collections.

Unlike traditional keyword searches, semantic retrieval attempts to represent the meaning of a query and compare it with the content or representation of scholarly documents. This can help researchers discover papers that use different terminology while addressing similar concepts.

Research on AI-assisted research workflows indicates that such technologies can reduce the time required for initial literature exploration. However, researchers must still verify the retrieved publications and assess their methodological and scholarly quality [2, 4].

### 2.2 Semantic Scholar

Semantic Scholar is an AI-supported academic search platform that combines scholarly metadata with semantic analysis and citation information. Its features can assist researchers in finding related publications and understanding citation relationships.

Its major strength is broad scholarly discovery through a combination of textual and citation-based information. This makes it potentially useful across disciplines, although the quality of results remains dependent on the underlying corpus and metadata.

### 2.3 Elicit

Elicit is designed around research questions rather than only conventional keyword queries. It can identify papers and organize information extracted from them into structured formats.

This approach is particularly relevant to literature reviews because researchers often need to compare studies according to variables such as research question, methodology, population, and findings. Nevertheless, AI-generated summaries should not be accepted without checking the original publications.

### 2.4 ResearchRabbit

ResearchRabbit emphasizes citation and author networks. Instead of treating literature discovery as a list of independent search results, it enables researchers to explore relationships between publications.

This network-oriented approach can be particularly useful after identifying a small number of high-quality “seed” papers. The researcher can then expand outward through related papers, authors, and citation connections.

### 2.5 Connected Papers and Litmaps

Connected Papers provides a visual representation of relationships among related publications. Litmaps similarly focuses on citation-based literature mapping and monitoring.

Such systems offer a different form of discovery from conventional search engines. Rather than simply ranking individual results, they allow researchers to examine the structure and development of a research area.

### 2.6 Cross-Disciplinary Considerations

The effectiveness of AI-powered discovery systems may vary between disciplines. Biomedical literature generally benefits from extensive indexing and standardized terminology. Computer science has a strong conference-publication culture, while social sciences and humanities contain more heterogeneous publication formats and terminology.

Recent work therefore emphasizes the importance of critically evaluating AI tools within the context of particular research practices rather than assuming that one technology is universally optimal [2, 4].

Table 1. Functional comparison of AI-powered literature discovery tools

| Tool | Primary approach | Main research use |
|---|---|---|
| Semantic Scholar | Semantic search and citation information | Finding relevant scholarly publications and exploring citation context |
| Elicit | Question-oriented search and evidence extraction | Comparing studies and organizing evidence for literature reviews |
| ResearchRabbit | Citation and author networks | Expanding discovery from seed papers and related authors |
| Connected Papers | Visual relationships among papers | Exploring related research and nearby papers |
| Litmaps | Citation mapping and monitoring | Tracking literature development and new publications |

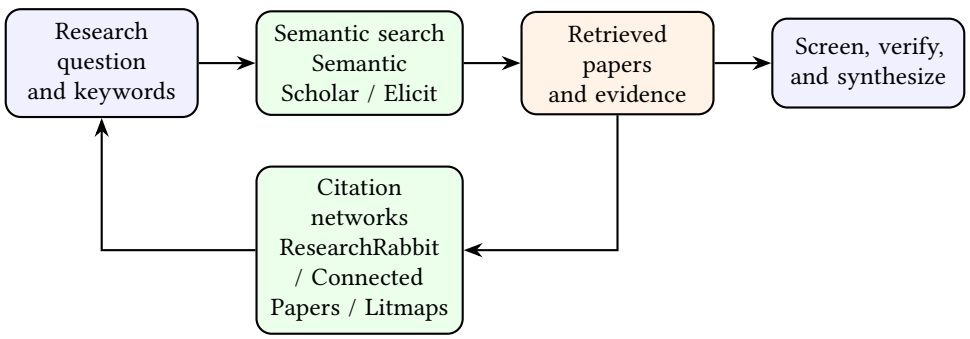



Fig. 1. Complementary roles of semantic search and citation-network tools in literature discovery.

## 3 Research Methodology

### 3.1 Research Design

This project uses a comparative evaluation design. Five AI-powered literature discovery tools are compared:

- Semantic Scholar
- Elicit

- ResearchRabbit
- Connected Papers
- Litmaps

Four disciplines are used as illustrative test domains:

(1) Computer science
(2) Biomedical science
(3) Social science
(4) Humanities

### 3.2 Search Topics

One research topic was selected for each discipline. The topics are shown in Table 2.

Table 2. Illustrative research topics by discipline

| Discipline | Illustrative research topic |
|---|---|
| Computer science | Explainable artificial intelligence |
| Biomedical science | Artificial intelligence in medical diagnosis |
| Social science | Social-media use and adolescent well-being |
| Humanities | Artificial intelligence and digital humanities |

Table 3. Cross-disciplinary test domains and proposed evaluation focus

| Discipline | Research topic | Proposed evaluation focus |
|---|---|---|
| Computer science | Explainable artificial intelligence | Coverage of conference and journal publications |
| Biomedical science | AI in medical diagnosis | Retrieval of relevant clinical and biomedical studies |
| Social science | Social-media use and adolescent well-being | Terminology variation and evidence relevance |
| Humanities | AI and digital humanities | Coverage of books, chapters, and heterogeneous publication formats |

The same research question within each domain should be entered into each tool as consistently as possible.

### 3.3 Evaluation Criteria

The tools were evaluated using six criteria:

- **Relevance:** How many retrieved publications directly addressed the research question?
- **Coverage:** How effectively did the tool identify important publications?
- **Novel discovery:** Did the tool identify useful publications that were not immediately apparent from conventional keyword searches?
- **Efficiency:** How much time was required to locate a useful collection of literature?
- **Usability:** How easy was the system to learn and operate?
- **Transparency:** Did the system provide sufficient information about why particular publications were recommended?

Table 4. Operational definitions of the evaluation criteria

| Criterion | Operational definition | Possible measure |
|---|---|---|
| Relevance | Proportion of retrieved publications judged relevant to the research question | Precision |
| Coverage | Proportion of the known relevant-paper set retrieved by a tool | Recall |
| Novel discovery | Useful relevant papers not identified by the predefined baseline search | Number of novel relevant papers |
| Efficiency | Time required to obtain a screened set of useful publications | Minutes per search task |
| Usability | Ease of completing the same predefined search tasks | Standardized user rating |
| Transparency | Clarity of the reasons, metadata, and source information shown for recommendations | Rubric-based score |

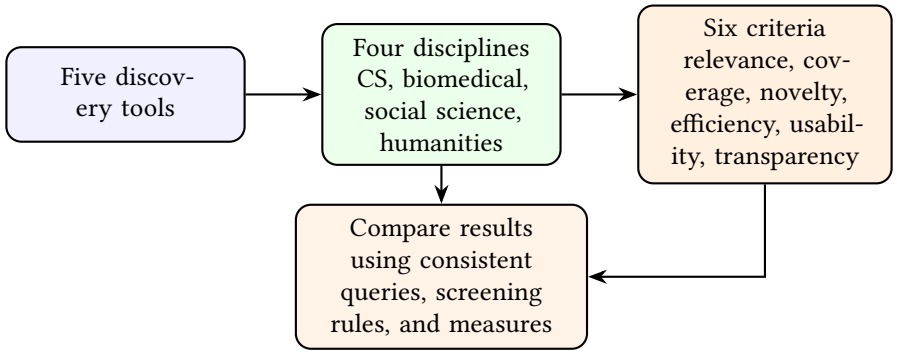



Fig. 2. Evaluation framework for comparing the five tools across four academic disciplines.

### 3.4 Quantitative Evaluation

Precision can be calculated using Equation 1:

$$\text{Precision} = \frac{\text{Relevant retrieved papers}}{\text{Total retrieved papers}} \tag{1}$$

Recall can be calculated using Equation 2:

$$\text{Recall} = \frac{\text{Relevant retrieved papers}}{\text{Total relevant papers}} \tag{2}$$

An overall performance score can then be calculated using predetermined weights. For example, Equation 3 defines an illustrative weighted score:

$$S = 0.25R + 0.20C + 0.15N + 0.15E + 0.15U + 0.10T \tag{3}$$

where $R$ is relevance, $C$ is coverage, $N$ is novel discovery, $E$ is efficiency, $U$ is usability, and $T$ is transparency. The weights are illustrative and should be justified or modified according to the objectives of the actual project.

## 4 Illustrative Results

**Important:** The following values are hypothetical examples intended to demonstrate how the results section could be written. They are not empirical measurements.

Table 5. Illustrative overall comparison

| Tool | Relevance | Coverage | Novel | Efficiency | Usability |
|---|---|---|---|---|---|
| Semantic Scholar | High | High | High | High | High |
| Elicit | High | Moderate to high | High | High | High |
| ResearchRabbit | Moderate to high | High | Very High | High | High |
| Connected Papers | Moderate to high | Moderate | Very High | High | High |
| Litmaps | Moderate to high | High | Very High | Moderate to high | High |

Table 6. Template for reporting measured results (replace dashes only after data collection)

| Tool | Discipline | Precision | Recall | Time (min) | Novel relevant papers |
|---|---|---|---|---|---|
| Semantic Scholar | Computer science | — | — | — | — |
| Elicit | Computer science | — | — | — | — |
| ResearchRabbit | Computer science | — | — | — | — |
| Connected Papers | Computer science | — | — | — | — |
| Litmaps | Computer science | — | — | — | — |

The illustrative results in Table 5 suggest that the tools have different strengths. Semantic Scholar performs strongly as a general-purpose discovery platform because it combines semantic search with citation information. Elicit is particularly useful when the research question requires structured comparison of papers.

ResearchRabbit, Connected Papers, and Litmaps appear particularly valuable for discovering relationships that may not emerge from a conventional keyword search. Their network-oriented interfaces allow researchers to move from one known publication to related studies, authors, and citation communities.

Table 7. Illustrative disciplinary performance

| Discipline | General-purpose tools | Network tools |
|---|---|---|
| Computer science | Semantic Scholar | ResearchRabbit |
| Biomedical science | Semantic Scholar / Elicit | Litmaps |
| Social science | Elicit / Semantic Scholar | ResearchRabbit |
| Humanities | ResearchRabbit / Semantic Scholar | Connected Papers / Litmaps |

These results should not be interpreted as proof that one tool is objectively superior. Rather, they demonstrate the expected importance of matching the discovery method to the characteristics of the discipline.

## 5 Discussion

The comparison suggests that AI-powered literature discovery is better understood as an ecosystem of complementary tools than as a competition to identify one universally superior platform.

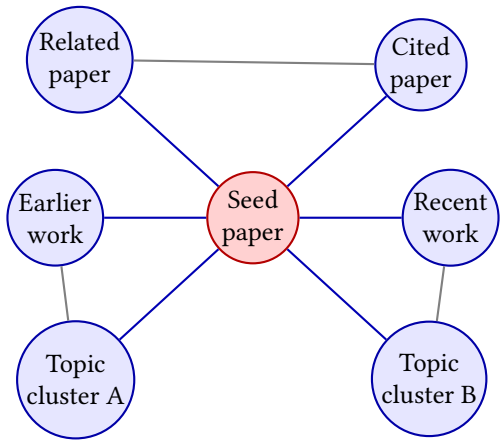



Fig. 3. Conceptual citation network showing how a seed paper can connect researchers to related, earlier, recent, and thematically clustered publications. This is an illustrative diagram, not a measured network.

Semantic search is useful during the initial stage of a project, when researchers may not yet know the terminology used by the field. Semantic Scholar and Elicit can therefore help researchers move beyond exact keyword matching.

Citation-network tools address a different problem. Once researchers have identified several important publications, ResearchRabbit, Connected Papers, and Litmaps can help reveal relationships between studies. This is especially valuable for identifying influential publications, research clusters, and developments within a field.

The cross-disciplinary dimension is particularly important. Academic disciplines do not share identical publication structures. Consequently, the effectiveness of a literature discovery system can depend partly on how well its underlying scholarly data represent a particular discipline.

For example, a system that performs strongly on journal articles may be less comprehensive in a field where books, book chapters, conference proceedings, or other publication formats are particularly important. Similarly, differences in terminology can influence semantic retrieval.

Another important issue is algorithmic bias. AI systems learn from available scholarly data and ranking mechanisms. If particular journals, languages, countries, institutions, or research communities are underrepresented in the underlying database, recommendations may reproduce those limitations.

Researchers should also distinguish between discovering a paper and evaluating a paper. An AI system may successfully identify a relevant publication, but this does not establish that the study is methodologically sound or that its conclusions are reliable. Human critical appraisal remains necessary [2].

Recent literature consequently emphasizes responsible and reflexive use of AI during literature reviews. AI systems should accelerate discovery while leaving interpretation, verification, and scholarly judgment with the researcher [2].

## 6 Advantages of AI-Powered Literature Discovery

Several advantages emerge from the comparison.

First, AI tools can reduce the time required for initial exploration. Researchers can rapidly identify related publications without constructing large numbers of keyword combinations.

Second, semantic retrieval can reveal publications using terminology that differs from the researcher's original query.

Third, citation-network visualization provides a useful method for understanding the structure of a research field.

Fourth, structured AI-assisted systems can support literature-review organization by helping researchers compare papers and extract potentially relevant information.

Finally, combining several tools can create a more comprehensive workflow than using a single search mechanism.

## 7 Limitations and Risks

### 7.1 Incomplete Coverage

No single scholarly platform contains the complete global research literature. Differences in indexing and metadata can therefore affect search results.

### 7.2 Algorithmic Bias

Ranking algorithms may favor highly cited or highly visible publications, potentially making less-cited but important research harder to discover.

### 7.3 Hallucination and Incorrect Information

Generative AI systems can occasionally produce inaccurate descriptions or references. Researchers should always verify bibliographic information against the original source [7].

### 7.4 Reproducibility

AI-powered ranking systems may change over time. A search performed today may not produce exactly the same results later.

### 7.5 Over-Reliance

Perhaps the greatest risk is treating AI recommendations as authoritative. AI should assist scholarly discovery rather than replace critical reading.

## 8 Recommended Workflow

Based on the comparison, a practical workflow would combine multiple approaches:

(1) **Initial search:** Use conventional academic databases and Semantic Scholar to establish a broad literature base.
(2) **Question-oriented discovery:** Use Elicit to explore research questions and identify potentially relevant evidence.
(3) **Citation expansion:** Use ResearchRabbit, Connected Papers, or Litmaps to identify related and influential publications.
(4) **Verification:** Check every important publication against the original paper and authoritative bibliographic databases.
(5) **Critical evaluation:** Assess methodology, evidence quality, limitations, and relevance independently of the AI recommendation.

This workflow treats AI as a research assistant rather than an autonomous replacement for scholarly judgment.

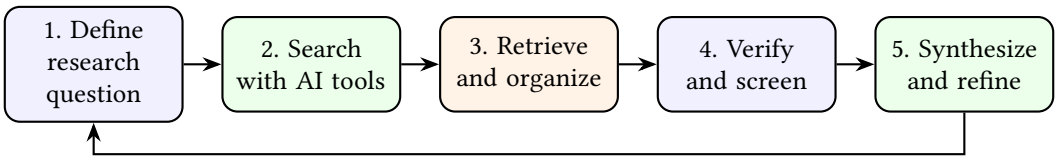



Fig. 4. Recommended workflow for combining AI-powered discovery tools with human verification and critical appraisal.

## 9 Conclusion

This project examined the potential of AI-powered literature discovery tools across multiple academic disciplines. The comparison demonstrates that different platforms address different aspects of scholarly discovery. Semantic Scholar and Elicit are well suited to semantic and question-oriented exploration, while ResearchRabbit, Connected Papers, and Litmaps provide valuable network-based approaches.

The main conclusion is that there is no universally optimal AI literature-discovery tool. Performance depends on the research task, discipline, database coverage, and characteristics of the literature being investigated.

AI-powered systems can substantially improve the efficiency of literature discovery, but their recommendations must be critically evaluated. The most reliable approach is therefore a hybrid workflow in which AI tools support searching and mapping while researchers retain responsibility for verification, interpretation, and scholarly judgment.

Future research should conduct larger controlled experiments using standardized benchmark datasets across more disciplines and languages. Such studies could provide stronger evidence regarding precision, recall, disciplinary bias, reproducibility, and the long-term effects of AI-assisted scholarly discovery.

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